LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART III

SESSION 12

Trade, Supply Chains & Industrial Policy



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Sáenz & Revilla, Creating More Resilient Supply Chains (MIT Sloan Management Review) ·
Facilitator: Bongo Adi & Kelechi Nnamdi

RECAP

Session 11 in Three Ideas

The trilemma

A country cannot have a fixed exchange rate, free capital flows and independent monetary policy all at once.

Capital flows

Hot money arrives in the good times and leaves in a hurry — a sudden stop hits reserves and the currency.

Mismatches

Borrowing in dollars while earning in naira turns a devaluation into a solvency problem.

TODAY'S PLAN

Session 12 at a Glance — 75 Minutes

Why countries trade, who Nigeria trades with, and how to build a supply chain that survives the next shock.



Recap & Hook

The AfCFTA opportunity — and why it is not automatic

Why Countries Trade

Comparative advantage and the gravity model

Supply Chains & Industrial Policy

De-risking, friend-shoring and the return of the state

Lab: Export Market Choice

Gravity, 'missing trade' and an AfCFTA scenario

BY THE END OF TODAY

Session 12 Learning Objectives

You will be able to:

01

Explain Why Trade Pays

Use comparative advantage to show why both sides gain — and who inside each country loses.

02

Size Export Markets

Use the gravity model to estimate expected trade and spot under-served African markets for AfCFTA.

03

Build Resilient Supply Chains

Measure supplier concentration and choose between diversifying, stockpiling and localising.

THE HOOK

A Market of 1.3 Billion People Next Door

“AfCFTA opens a continent. Let's export everywhere next year.”

Before the board approves an export budget, ask three questions:

Can we get there?

Distance, roads, ports and border delays often cost more than the tariff the agreement removes.

Is anyone buying?

Market size matters: a rich, nearby market beats a poor, distant one — that is the gravity model.

Is our own supply chain safe?

Exporting more means importing more inputs. If one supplier country stops, so do we.

NIGERIAN BUSINESS CASE

AfCFTA and a Nigerian Manufacturer

The situation

A Lagos packaged-food manufacturer sells only in Nigeria. AfCFTA lowers tariffs on most goods traded between member states, and the board wants an export plan. Most of its packaging and machinery are imported.

The decision question

Which **African markets** should it enter first — and how should it make its **supply chain** shock-proof before it grows?

The data we have

ng_trade_partners.csv — Nigeria's exports and imports with 30 partners, plus partner GDP, distance, language, AfCFTA membership and tariffs. *Stylised teaching data.*

What a good answer looks like

A ranked shortlist of **three export markets** with reasons, and **two actions** to reduce supply-chain concentration.

PART

01

Why Countries Trade

Specialise in what you do relatively best.

Everyone Gains From Doing What They Do Relatively Best

IN PLAIN ENGLISH

A country gains from trade by producing what it gives up **least** to make — its **comparative** advantage — and buying the rest. Even a country that is better at everything gains by specialising.

EXAMPLE

If a Nigerian worker can make 10 bags of cement or 2 phones a day, and a Chinese worker 12 bags or 6 phones, Nigeria gives up 0.2 phones per bag; China gives up 0.5. Nigeria has the comparative advantage in **cement**, China in **phones** — even though China is better at both.

THE FORMULA

$$\text{opportunity cost of X} = \text{output of Y given up} \div \text{extra output of X}$$

WHY A MANAGER CARES

Comparative advantage is not fixed. Nigeria moved from importing most of its cement to producing almost all of it once investment, gas and scale came together. Your firm's export opportunity lies where your **cost relative to rivals** is lowest.

Trade Raises the Total — Not Everyone's Share

Who gains

Usually many, spread thinly

- Consumers: cheaper and more varied goods
- Exporters: bigger markets and scale
- Firms buying cheaper imported inputs
- The economy as a whole: higher income

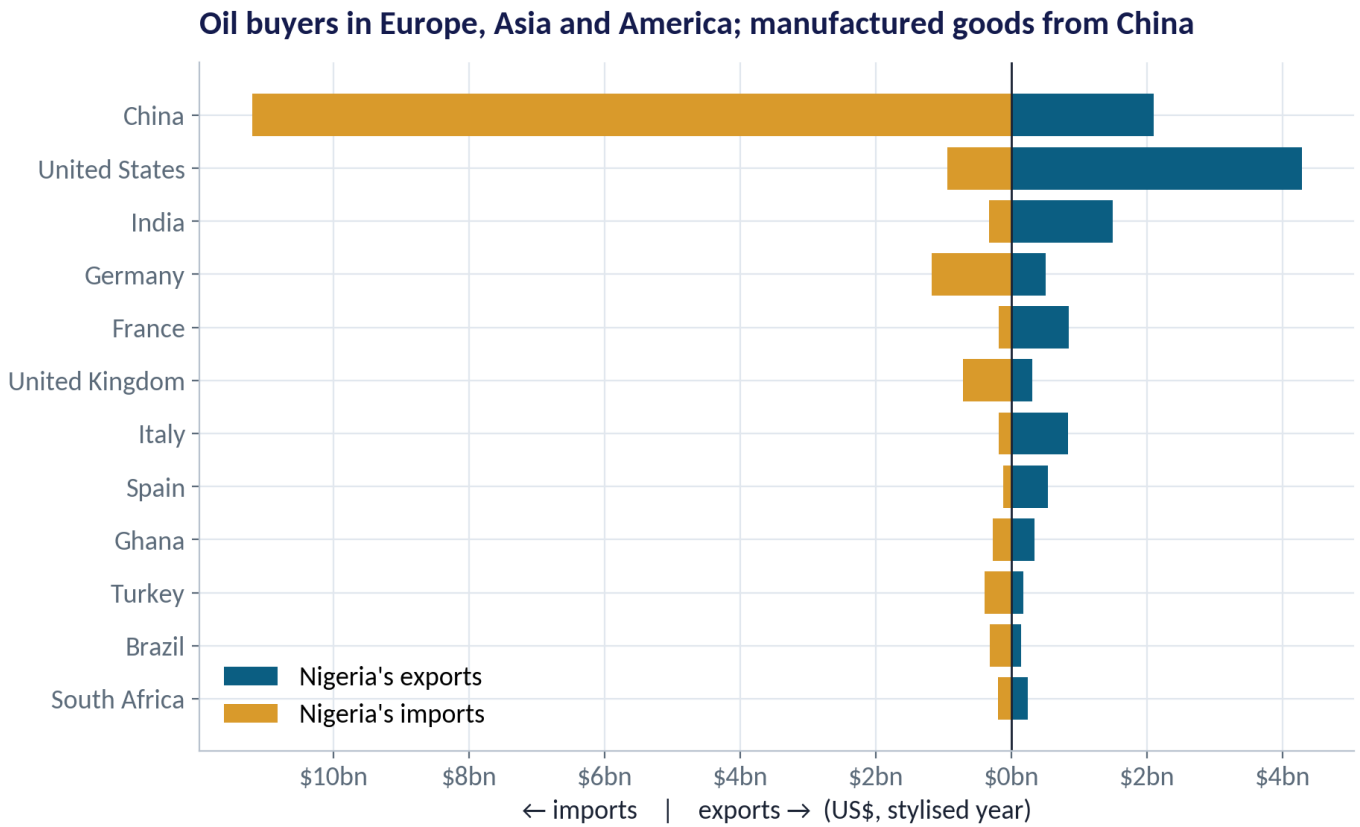
Who can lose

Usually few, hit hard

- Firms competing directly with imports
- Workers in those firms and towns
- Producers who relied on protection
- These losers lobby — which is why trade is political

*Good trade policy grows the pie **and** helps the losers adjust. Ignoring the losers is how trade agreements lose public support.*

Who Nigeria Trades With



Source: EEGB stylised teaching data (data/). Not official statistics.

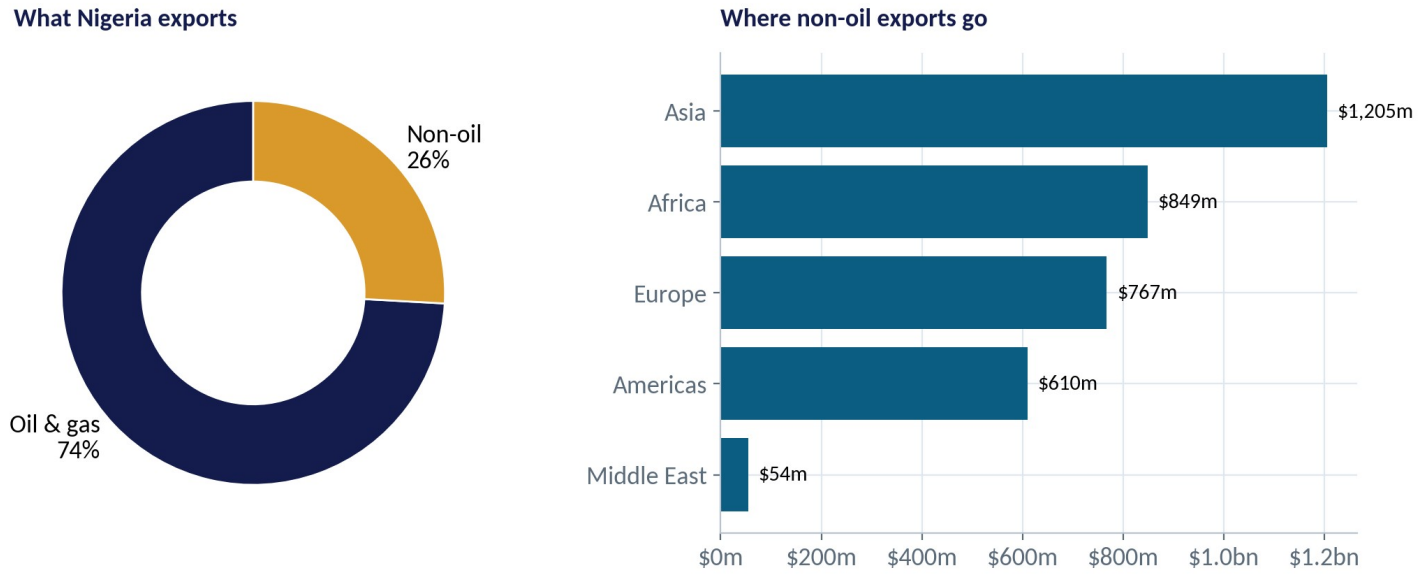
Fig S12-F01 · replicate: s12_figures.R / .py

WHAT TO SEE

- Exports go mainly to **oil buyers**: the United States, India and European refiners.
- Imports come overwhelmingly from **China**: machinery, phones, textiles, inputs.
- Total trade in this stylised year is about **$31bn**; trade with African partners is only **7.6%** of it.
- That low African share is the gap AfCFTA is meant to close.

Export Concentration: Oil Still Dominates

One commodity dominates exports — Africa is already the No. 2 non-oil market



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S12-F05 · replicate: s12_figures.R / .py

WHAT TO SEE

- Oil and gas make up about **74%** of exports in these data.
- When oil prices or output fall, export earnings — and dollars for importers — fall with them.
- Non-oil exports are about **$3.5bn**; Africa already takes **$849m** of them, second only to Asia.
- Manufactured and processed goods are more likely to sell in Africa than in Europe.

Trade Behaves Like Gravity

IN PLAIN ENGLISH

Countries trade more with partners that are **big** and **close**, and less with those that are small and far — just as gravity pulls more strongly between heavy, nearby objects. Shared language, borders and history add to the pull; being landlocked subtracts.

EXAMPLE

In our data b_1 = **0.98** and b_2 = **−1.08**: a partner 10% bigger trades about 10% more with Nigeria; one 10% further away trades about 11% less.

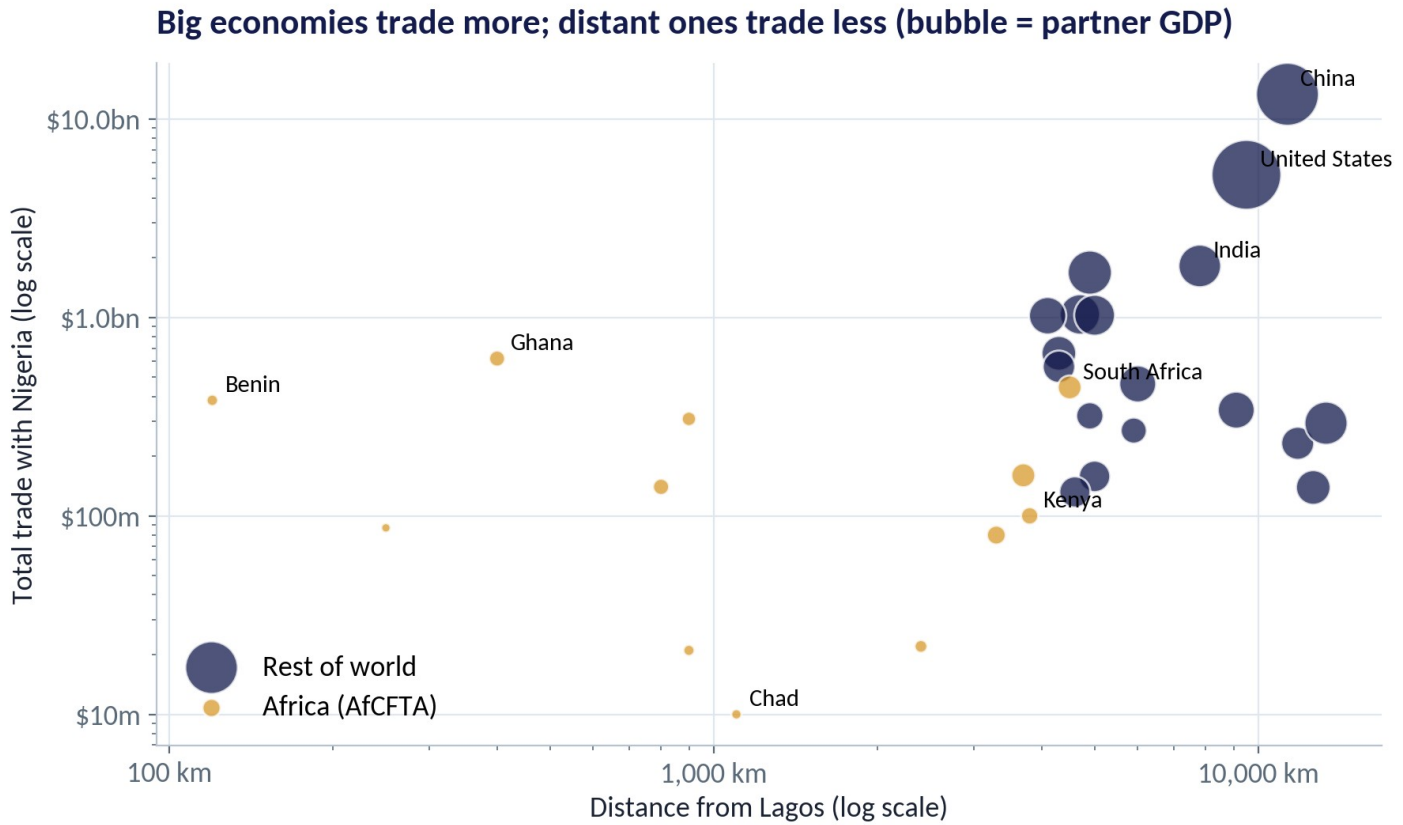
THE FORMULA

$$\log(\text{trade}) = a + b_1 \cdot \log(\text{GDP}) + b_2 \cdot \log(\text{distance}) + \text{extras}$$

WHY A MANAGER CARES

Gravity is the most reliable model in international economics. It tells you what trade **should** be with each market — so you can spot where trade is **missing** and where the opportunity is real.

Size Pulls, Distance Pushes



Source: EEGB stylised teaching data (data/). Not official statistics.

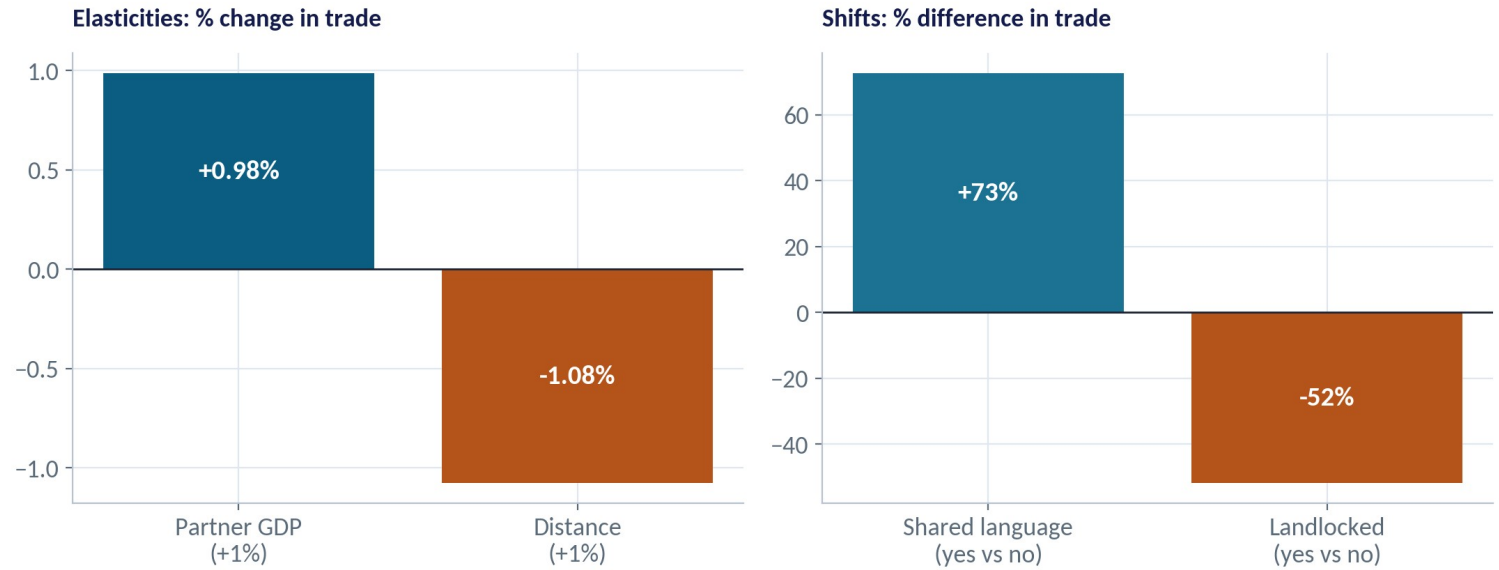
Fig S12-F02 · replicate: s12_figures.R / .py

WHAT TO SEE

- Each bubble is a partner; bigger bubble = bigger economy.
- Distant giants (China, the US) still trade heavily with Nigeria because of their size.
- Neighbours like **Benin and Ghana** trade a lot for their size — they are very close.
- Small, landlocked and distant partners such as **Chad** trade very little.

What Drives Nigeria's Trade?

Gravity model of Nigeria's trade (R^2 = 0.89): size pulls, distance pushes



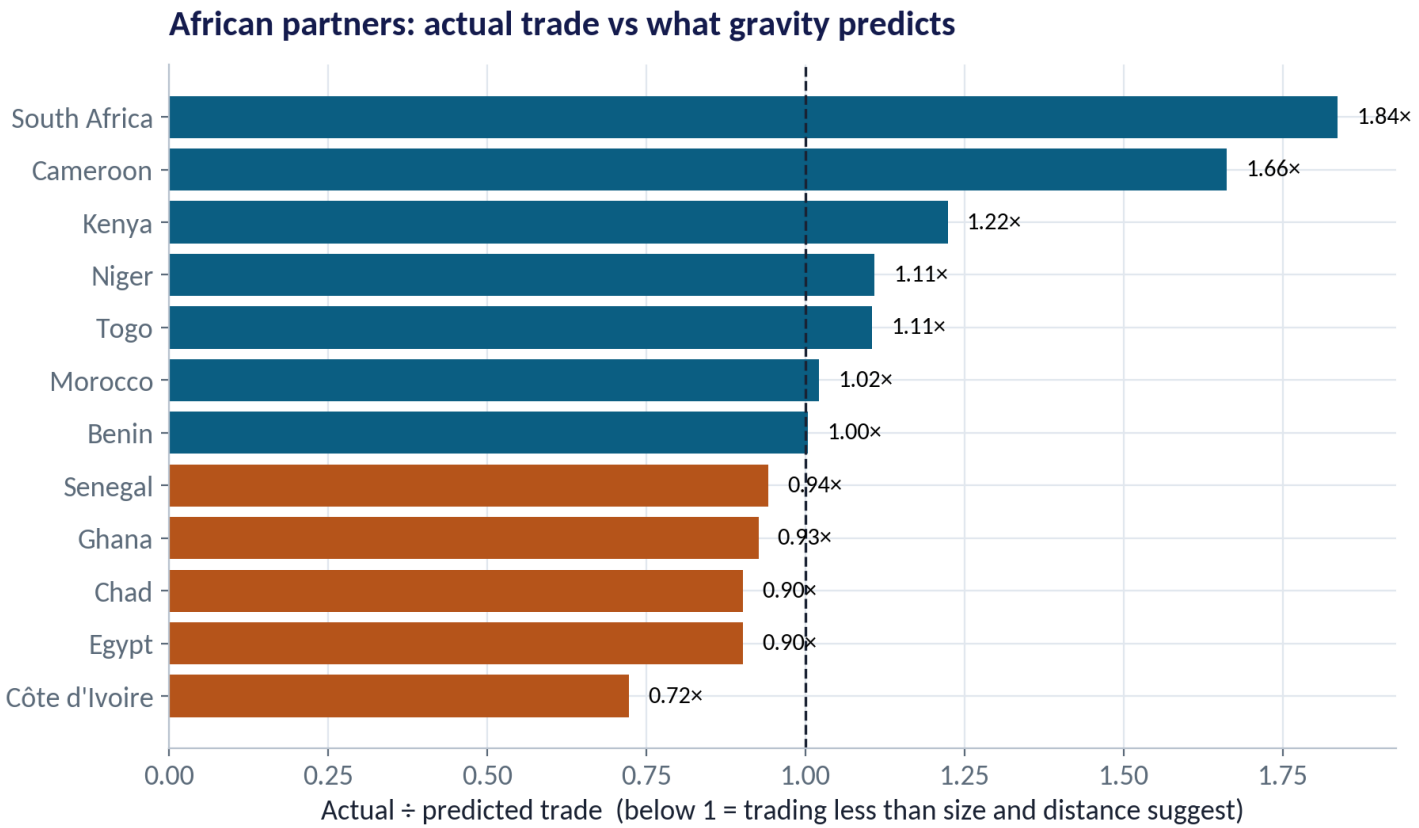
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S12-F03 · replicate: s12_figures.R / .py

WHAT TO SEE

- **GDP** elasticity ≈ **0.98**; **distance** elasticity ≈ **−1.08**.
- A **shared language** goes with about **73%** more trade.
- **Landlocked** partners trade about **52%** less — every border and truck adds cost.
- Four variables explain **89%** of the variation (R^2 = 0.89).

Where Is Trade Below What Gravity Predicts?



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S12-F04 · replicate: s12_figures.R / .py

WHAT TO SEE

- A ratio below 1 means Nigeria trades **less** than size and distance suggest.
- **Côte d'Ivoire (0.72×)**, Egypt and Ghana sit below the line — potential room to grow.
- South Africa (1.84×) and Cameroon (1.66×) already trade more than predicted.
- Missing trade is a **clue**, not a verdict: check non-tariff barriers, payments and competition first.

Quick Check: What Would You Say?

?

The gravity model says Nigeria trades only 0.72 times the predicted amount with Côte d'Ivoire. What is the best interpretation for an exporter?

- **A** Ivorians do not like Nigerian products
- **B** There may be unexploited potential — investigate the barriers (language, logistics, payments, rules) before investing
- **C** The model is wrong and should be ignored
- **D** Côte d'Ivoire is too far away to matter

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

AfCFTA: Promise and Practice

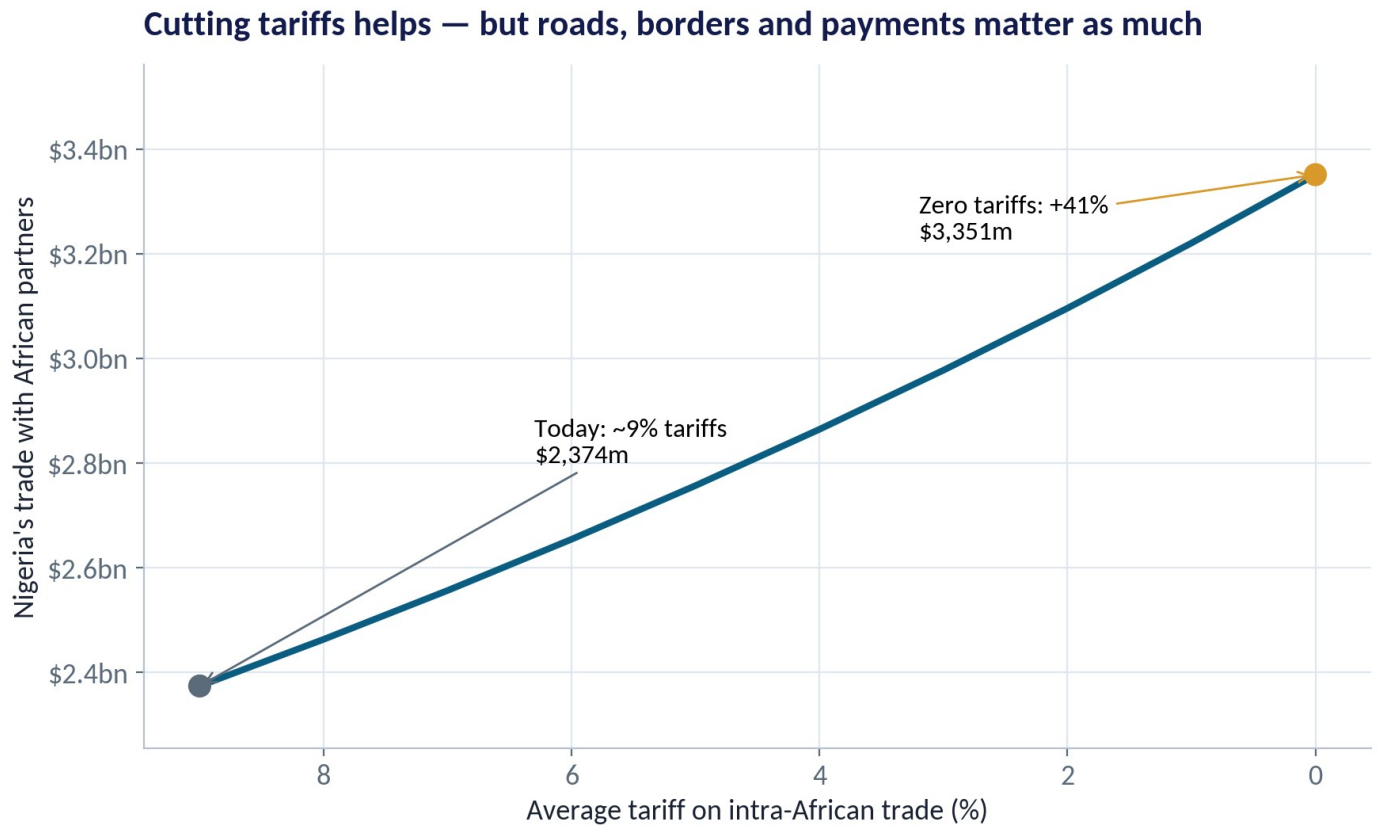
Tariffs are only part of the cost of trading.

The African Continental Free Trade Area

Feature	What it means for a Nigerian firm
Coverage	Nearly all African Union members have signed; Nigeria signed in 2019 and ratified in 2020. Trading under the agreement began in January 2021.
Tariffs	Most tariff lines are phased down to zero over several years; sensitive and excluded lists remain.
Rules of origin	Goods must be 'made in Africa' to qualify — assembling imported kits may not count.
Beyond goods	Protocols on services, investment, competition and digital trade are being negotiated and implemented in stages.
Payments	The Pan-African Payment and Settlement System (PAPSS) aims to let firms pay in local currencies instead of dollars.

Implementation is gradual and uneven. Check the current schedules for your product before you plan prices.

What Removing Tariffs Could Do



Illustrative: trade proportional to (1 + tariff)^-4. Source: EEGB stylised teaching data. Not official statistics.

Fig S12-F07 · replicate: s12_figures.R / .py

WHAT TO SEE

- Trade with African partners: about **$2.4bn** today in these data.
- If average tariffs fell from **9% to zero** and trade responds with an elasticity of 4, trade rises about **41%** to **$3.4bn**.
- The elasticity is uncertain — estimates often range from 3 to 6 — so treat this as a range, not a forecast.
- Poor roads, slow borders, checkpoints and payment frictions can cost more than the tariff itself.

How Firms Get AfCFTA Wrong

THE TRAP	THE BETTER HABIT
✗ “Tariffs are gone, so our prices will be competitive.”	✓ Add up **landed cost**: freight, border delays, informal payments, FX conversion and distribution.
✗ Assuming every product qualifies.	✓ Check **rules of origin** and the tariff schedule for your exact product code.
✗ Choosing markets by population alone.	✓ Use **gravity**: income, distance, language and existing trade links.
✗ Ignoring currency: pricing in naira for CFA or cedi markets.	✓ Plan **invoicing currency** and FX risk; explore local-currency payment rails.
✗ Going everywhere at once.	✓ Start with **two or three** markets, learn, then expand along trade corridors.

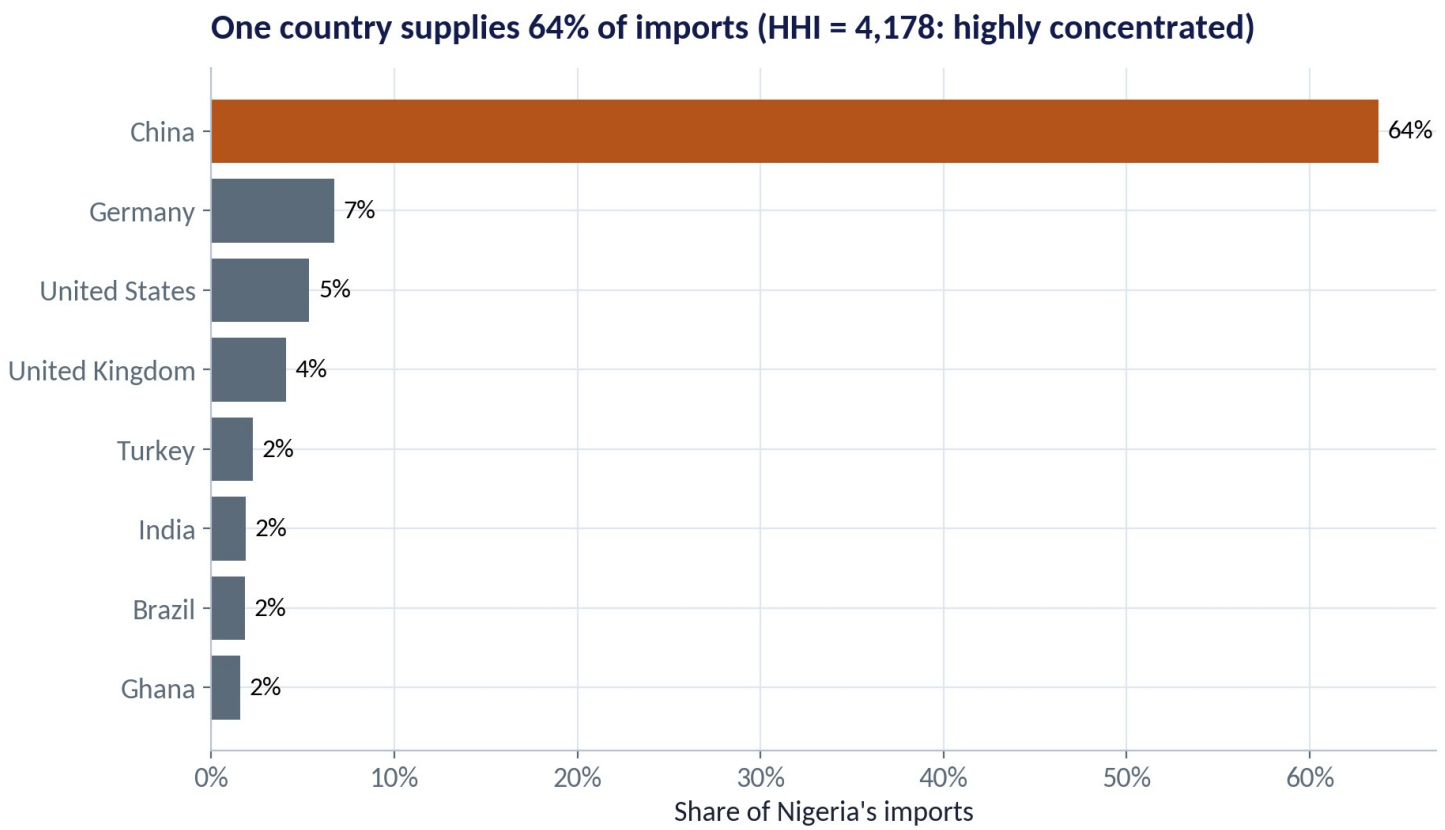
PART

03

Supply Chains and Industrial Policy

Efficient was yesterday's goal. Resilient is today's.

One Supplier Country, One Point of Failure



Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- China supplies about **64%** of Nigeria's imports in these data.
- The **Herfindahl–Hirschman Index** (sum of squared shares) is **4,178** — above 2,500 is commonly treated as highly concentrated.
- The top three sources supply **76%** of imports.
- A port closure, export ban or freight spike in one place can stop Nigerian factories.

The Herfindahl–Hirschman Index

IN PLAIN ENGLISH

Square each supplier's percentage share and add them up. **One supplier = 10,000**. Many small suppliers = close to zero. The higher the number, the more your supply depends on a few sources.

EXAMPLE

64% from China, 7% from Germany, 5% from the US and many small sources → HHI ≈ **4,178**. Splitting China's share across two countries of 32% each would roughly halve that contribution.

THE FORMULA

$$HHI = s_1^2 + s_2^2 + \ldots + s_n^2 \quad \text{(shares in \%)}$$

WHY A MANAGER CARES

Run this on **your own** purchase ledger by country and by supplier. An HHI above 2,500 on a critical input is a board-level risk.

Four Ways to Make a Supply Chain Shock-Proof

Diversify suppliers

Qualify a second source in another country (e.g. Turkey or India alongside China). Costs more per unit, but buys insurance.

Hold strategic stock

Keep more inventory of critical, slow-to-replace inputs. Ties up cash — expensive when interest rates are 25%+.

Localise or regionalise

Source within Nigeria or AfCFTA partners. Shorter routes and no FX for intra-Nigerian supply, but capability takes time.

See the whole chain

Map tier-2 and tier-3 suppliers. Many 'diversified' firms discover their suppliers all depend on the same factory.

*Sáenz and Revilla's core point: resilience must be **designed in** — visibility, flexibility and collaboration — not bolted on after a crisis.*

De-risking, Friend-Shoring and Industrial Policy

Trend	What it is	Implication for Nigeria
De-risking	Reducing dependence on one country without cutting trade off entirely	Western buyers may look for alternative suppliers — an opening for African producers
Friend-shoring	Moving supply chains to politically aligned countries	Alignment and stability start to matter as much as cost
Export controls	Restrictions on selling strategic goods (e.g. advanced chips)	Technology access can be cut by decisions taken far away
Industrial policy	Subsidies and rules to build strategic industries (e.g. US CHIPS Act and Inflation Reduction Act)	Rich-country subsidies compete for the same investment Nigeria wants
Local content & backward integration	Policies requiring or rewarding local production	Can build industries (cement) — or protect inefficiency if never phased out

When Does Industrial Policy Work?

✓ More likely to work

Discipline and competition

- Support tied to **exports** or clear performance targets
- Time-limited, with a sunset clause
- Builds on real comparative advantage (gas, agriculture, scale)
- Firms still face competition

✗ More likely to fail

Protection without discipline

- Import bans that last indefinitely
- Support captured by a few connected firms
- Goes against cost reality (no inputs, no power)
- No measure of success

*Ask of any policy: what would make it **end**, and how would we know it worked?*

CONCEPT CHECK

Quick Check: What Would You Say?

?

Your firm imports 70% of a critical packaging input from one country. Interest rates are 27%. Which response is most sensible first?

A. Triple inventory immediately

B. Qualify a second supplier in another country and map who your current supplier depends on

C. Stop importing and wait for local supply to appear

D. Do nothing — diversification is too expensive

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Where Should We Export First?

YOU WILL NEED

Data
`data/ng_trade_partners.csv`

R script
`sessions/s12/R/s12_lab.R`

Python script
`sessions/s12/python/s12_lab.py`

⏱ **20 minutes**

1. Rank partners by **total trade** and compute Africa's share.
2. Fit the **gravity model**: log trade on log GDP, log distance, language and landlocked.
3. Compare **actual vs predicted** trade for each African partner.
4. Run the **AfCFTA tariff scenario** with elasticities of 3, 4 and 6.
5. Compute the **HHI** of import sources and propose two de-risking actions.

Deliverable: a ranked shortlist of three export markets and two supply-chain actions, one line of reasoning each.

WORDS TO OWN

Today's Vocabulary in Plain English

Comparative advantage	Producing what you give up least to make, relative to others.
Opportunity cost	What you give up to produce or buy something.
Gravity model	Trade rises with partner size and falls with distance.
Elasticity	% change in one thing for a 1% change in another.
AfCFTA	The African Continental Free Trade Area.
Rules of origin	Rules on how much of a good must be made locally to qualify.
Non-tariff barrier	Any cost of trading other than a tariff: delays, permits, quotas.
HHI	Sum of squared shares; measures concentration.
Friend-shoring	Moving supply chains to politically aligned countries.
Industrial policy	Government support to build particular industries.

KEY TAKEAWAYS

Session 12 in 5 Ideas

01 Countries — and firms — gain by specialising in their **comparative advantage**.

02 Trade follows **gravity**: big and close partners matter most; use it to spot **missing trade**.

03 AfCFTA cuts tariffs, but **logistics, rules of origin, payments and FX** decide whether you can compete.

04 Measure supplier **concentration**; diversify, stock, localise and map deeper tiers.

05 Industrial policy works best with **discipline, competition and an exit date**.



NEXT › Session 13: Climate & Energy Transition

Before next session: Read Stern, The Economics of Climate Change (summary chapters) and skim the KPMG / IFRS Foundation guide on ISSB and TCFD. List your company's three largest sources of emissions.